

Math 290

Practice Quiz 2 (Original from Canvas) — Solutions

Short-response key: 1(a) A, 1(b) True, 1(c) False.

1(a) Answer: A. The theorem is true and the proof is correct. The two cases “ x odd” and “ x even” cover every integer x , so no separate case for y is needed (that rules out E). Assuming x is odd or even *inside a case* is exactly how proof by cases works (that rules out C and D). In Case 1 the algebra $y = (2k + 1) - (2\ell + 1) = 2(k - \ell)$ correctly shows y is even, so $xy = 2((k - \ell)x)$ is even. (Based on Section 8.A.)

1(b) Answer: True. If $a \equiv 3 \pmod{5}$, then $a = 5k + 3$ for some $k \in \mathbb{Z}$, so

$$a^2 = 25k^2 + 30k + 9 = 5(5k^2 + 6k + 1) + 4.$$

Thus $a^2 - 4 = 5(5k^2 + 6k + 1)$, so $5 \mid (a^2 - 4)$, i.e. $a^2 \equiv 4 \pmod{5}$. (Based on Definition 8.11 and Proposition 8.13.)

1(c) Answer: False. The converse of “If x is even, then $x + 1$ is odd” is “If $x + 1$ is odd, then x is even.” The given sentence, “If $x + 1$ is even, then x is odd,” is $\neg Q \Rightarrow \neg P$, which is the *contrapositive*. (Based on Section 7.C.)

2. (Based on Section 7 (contrapositive) and Section 8 (cases).)

Solution. Let $x, y \in \mathbb{Z}$. We work contrapositively. The negation of “ x is odd and y is even” is “ x is even or y is odd” (De Morgan). So assume x is even or y is odd. Note that $x^2 - xy = x(x - y)$.

Case 1: x is even. Then $x = 2k$ for some $k \in \mathbb{Z}$, so $x^2 - xy = 2(k(x - y))$. Since $k(x - y) \in \mathbb{Z}$, this is even.

Case 2: x is odd. Then x is not even, so by our assumption y is odd. Write $x = 2k + 1$ and $y = 2\ell + 1$ with $k, \ell \in \mathbb{Z}$. Then $x - y = 2(k - \ell)$, so

$$x^2 - xy = x(x - y) = 2(x(k - \ell)),$$

which is even since $x(k - \ell) \in \mathbb{Z}$.

In every case $x^2 - xy$ is even. This proves the contrapositive, so the original implication holds. $\square$

Alternative (direct): if $x(x - y)$ is odd, then x and $x - y$ are both odd (otherwise the product would be even, as in Proposition 8.5). Then $y = x - (x - y)$ is odd minus odd, which is even.

3. (Based on Section 9.C and Exercises 9.6–9.7.)

Solution. Let $x, y, z \in \mathbb{R}$ with $x, y \in \mathbb{Q}$, $x \neq 0$, and $z \in \mathbb{R} - \mathbb{Q}$. (Since $x \neq 0$, the number $\frac{y+z}{x}$ is defined.) Assume, by way of contradiction, that $\frac{y+z}{x}$ is rational. Write

$$x = \frac{a}{b}, \quad y = \frac{c}{d}, \quad \frac{y+z}{x} = \frac{e}{f}$$

with $a, b, c, d, e, f \in \mathbb{Z}$ and $b, d, f \neq 0$. Multiplying by x and subtracting y ,

$$z = \frac{e}{f} \cdot \frac{a}{b} - \frac{c}{d} = \frac{ae}{bf} - \frac{c}{d} = \frac{ade - bcf}{bdf}.$$

Here $ade - bcf \in \mathbb{Z}$ and $bdf \in \mathbb{Z}$ with $bdf \neq 0$ (a product of nonzero integers). So $z \in \mathbb{Q}$, which contradicts the assumption that z is irrational. Therefore $\frac{y+z}{x}$ is irrational. $\square$

4. (Based on Section 10.)

Solution. Let S and T be sets.

($\Rightarrow$) Assume $S \cap T = S$. Let $x \in S$. Then $x \in S \cap T$ (substitution, using the assumption $S = S \cap T$). So $x \in S$ and $x \in T$ (definition of intersection). Hence $x \in T$ (simplification: $P \wedge Q \Rightarrow Q$). Therefore $S \subseteq T$ (definition of subset).

($\Leftarrow$) Assume $S \subseteq T$.

First, let $x \in S \cap T$. Then $x \in S$ and $x \in T$ (definition of intersection). Hence $x \in S$ (simplification: $P \wedge Q \Rightarrow P$). Therefore $S \cap T \subseteq S$ (definition of subset).

Next, let $x \in S$. Then $x \in T$ (definition of subset, since $S \subseteq T$). So $x \in S$ and $x \in T$ (conjunction of the two facts). Hence $x \in S \cap T$ (definition of intersection). Therefore $S \subseteq S \cap T$ (definition of subset).

Since $S \cap T \subseteq S$ and $S \subseteq S \cap T$, we have $S \cap T = S$ (definition of set equality). $\square$